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MEE342: Phase 1 Report

System description and functional overview

As a group we have chosen to replicate the system of a transfer case. The function of a transfer case is to distribute the engine's power to both the front and rear axles of a vehicle. This system is commonly called 4-Wheel Drive. A system like this is commonly found in SUV's or pickup trucks where they are commonly found in rough terrain and must be able to obtain traction from all 4 tires in order to maneuver through the terrain.

Traditionally in a 2WD vehicle if you were stuck in mud and stepped on the gas, there is a high possibility that your rear wheels do not provide enough traction to maneuver the vehicle out. In 4WD you engage all 4 tires of the vehicle to spin at the same speed which allows you to crawl out. This same scenario applies to different conditions such as snow, sand, or gravel. In some applications there is a 4WD-High and 4WD-Low. In 4WD-Low you essentially sacrifice speed for torque by having a gear reduction. This reduction allows you to multiply your torque but cannot go faster than a specific speed without causing damage to the transfer case. In some instances of a transfer case you will find a clutch assembly however it is not required in all circumstances.

Subsystem breakdown and Components

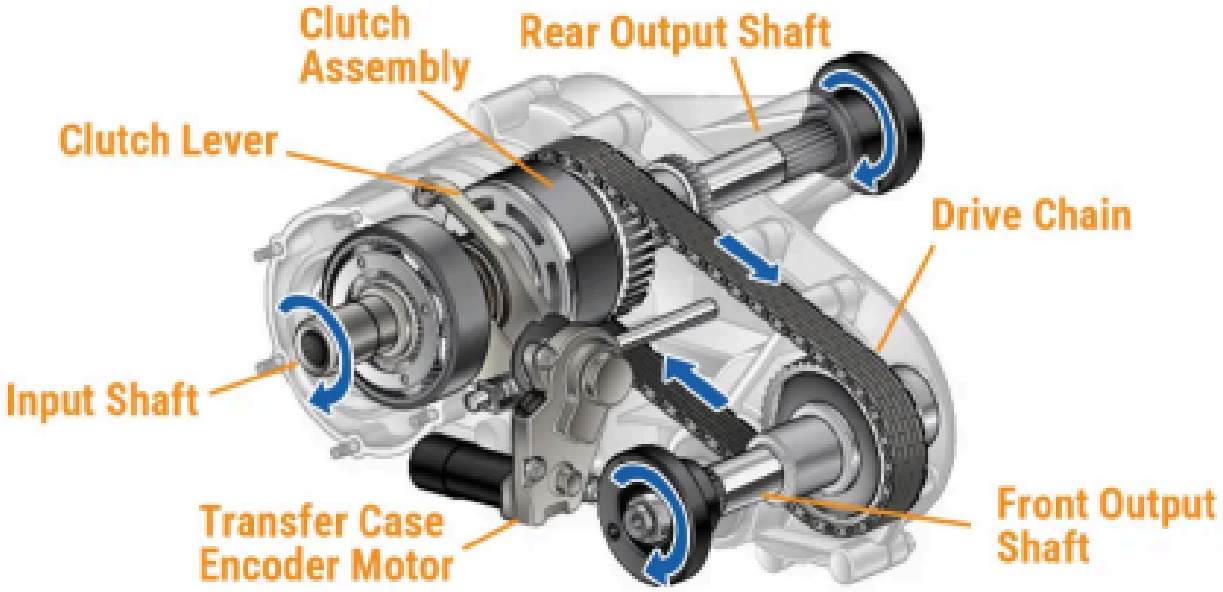
In a system like this we can break down the system into 3 subsystems; Housing, Geartrain and Shifting System

Housing consists of the cases which are the outer metal shells which hold all the components. Inside the cases we have gaskets and seals to ensure a minimal gap between other components to prevent fluids from leaking. This transfer case also stores the oil that will lubricate and cool and components to mitigate stress failure.

The geartrain is how you will transfer the power from the engine to the front and rear wheels. To do this you must have the Input shaft and gears, this is essentially where the power transfer begins. To ensure maximum efficiency there are bearings installed that hold the shafts and prevent as much friction as possible.

The shifting system can be either a lever or an electrical motor that shifts the transfer

case into different positions. This is essentially what changes your vehicle from 2WD to 4WD. In recent years there has been more of a shift to electrical motors however in most vehicles you will see a lever. This lever is pulled which in response grabs the shift works forward or backward. This allows the transfer case to shift from 2WD to 4WD.



Preliminary failure mode review

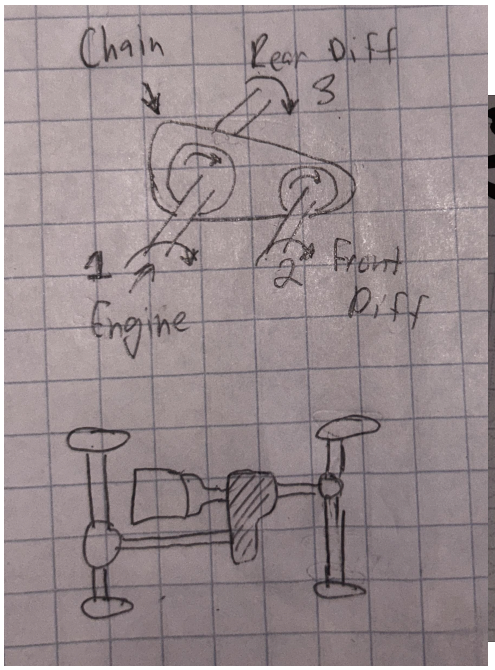
The main failure involving a transfer case is misalignment and fatigue. The misalignment becomes an issue when dealing with gear reduction because a simple miscalculation could result in slipping, power loss, and catastrophic failure. Our design accounts for this issue by ensuring that all shafts are perfectly in line, and the gears are created with the assistance of design software that specializes in these cases.

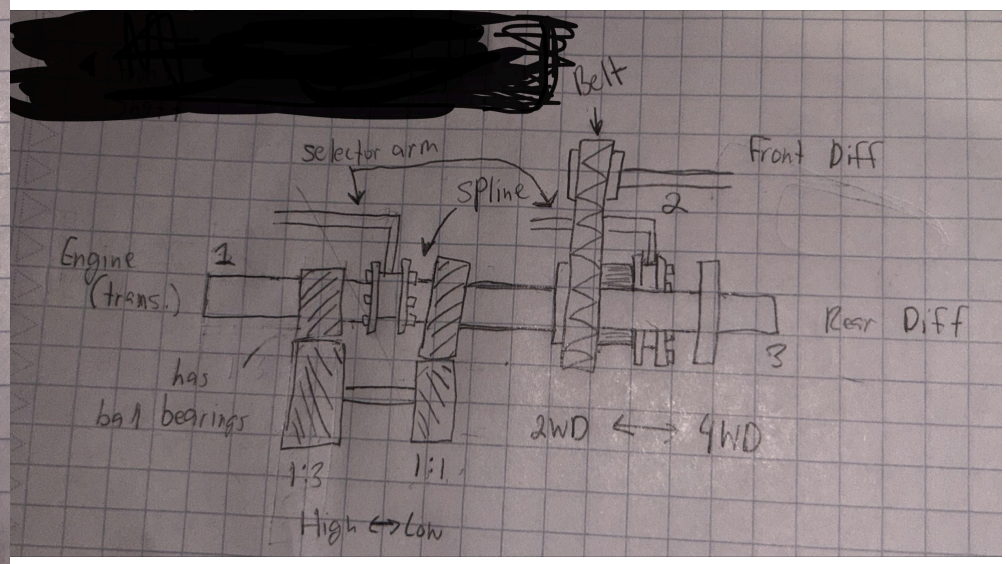
Fatigue is an issue in full-size transfer cases because they are designed to withstand heavy loads for prolonged amounts of time with failure points built in so that the maintenance covers most potential issues with wear. Our design fatigue will come from the testing of the 3d printer filament and determining the speed at which we can run a motor on it without damaging the internal components. This is also a common practice for full-size transfer cases, as it is a common rule of thumb to never exceed 40mph while in 4HI and 5mph while in 4LOW.

Critical design parameters

- Power transmission
    - Belt/ sprocket tooth count: defines the gear ratio and torque distribution
    - Center-to-center distance from pulley to front output shaft
- Shaft and axle parameters
    - Shaft diameters: affect torsional strength, bending resistance
    - Shaft lengths: determine the size of the housing case
- Bearing/ support parameters
    - Bearing type: must be compatible with shafts and gears
    - Bearing inner/outer diameter
    - Clearance for 3D printed parts
- Housing
    - Housing wall thickness: affects weight and rigidity
    - Internal clearances: prevent locking up
- Material and manufacturing
    - Filament type: affects strength, flexibility, fatigue
    - Infill and print settings: affect material cost, print time, and strength
    - Material tensile and compressive strength limits
- General and Manufacturing Parameters
    - Overall gear ratio: derived from tooth counts which affects 4WD efficiency
    - Alignment tolerances for smooth operation and reduce vibration
    - Fastener types and sizes: for assembling printed parts
    - Thermal expansion coefficients: If mixing materials, to include temperature and misalignments.
    - Safety factors for loads: Applied to all parameters to handle max torque or impacts in 4WD scenarios

Kinematics





ω_d = angular speed of the drive pulley

ω_f = angular speed of the drive pulley

N_d, N_f = belt pulley tooth counts

<u>belt speed</u>

$$\omega_f = \omega_d \left(\frac{D_d}{D_f}\right)$$

<u>tooth counts</u>

$$\omega_f = \omega_d \left(\frac{N_d}{N_f}\right)$$

<u>torque</u>

$$T_f = T_d \left(\frac{D_f}{D_d}\right) \eta$$

η ~ belt efficiency